

Alexandre D'Hooge

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Open to postdoctoral positions in Music Information Retrieval.

Education

Algomus Team, CRISAL Laboratory, Université de Lille <i>PhD Thesis - Supervised by Mathieu Giraud, Louis Bigo, and Ken Déguernel.</i> Assisting Western Popular Music Guitar Practice and Tablature Composition with Machine Learning.	2022-2025
IRCAM, France <i>ATIAM Master's Degree</i> Master's degree in acoustics, signal processing and computer science for music. Introduction to musicology, audio perception and audio production tools (PureData, Ableton, MaxMSP).	2021-2022
ENS Paris-Saclay, France <i>M2FESup Intranet - Master's degree in computer engineering</i> Intensive one-year class preparing for the French examination <i>Agrégation</i> which I passed ranking 3rd (out of 15).	2019-2020
<i>Première année de Master - First year of master's degree in electronics</i> Digital and analog electronics, automation engineering, signal processing, electrical engineering, image processing, digital filtering and transmission, non-linear automatics, source encoding	2018-2019
<i>L3 SAPHIRE - Bachelor's degree in applied physics for engineering, research and education</i> Solid mechanics, fluid mechanics, digital and analog electronics, computer science, automation engineering, quantum mechanics.	2017-2018

Work Experience

Teaching

2022-2025

Teaching Assistant - Université de Lille

- Object Oriented Programming in Java (2nd year Bachelor).
- Introduction to Unix (2nd year Bachelor).
- Automata and Regular Languages (2nd year Bachelor).
- Computer Science Project in Java (2nd year Bachelor).
- Internship tutoring (3rd year Bachelor).

Supervision Activities - Université de Lille

- Assistance in supervising a Master's Thesis in Musicology on Brutal Death Metal.
- Main Supervisor of a Master's Project on Conditional Bass Tablature Generation.
- Co-Supervisor of a Master's Student internship (6mo) on Assisted Guitar Teaching.
- Supervision of a group of high-school interns visiting the lab for 1 week.
- Co-Supervisor of a Bachelor Student internship.

Research Internships

Research Visit (during the 2nd year of my PhD)

Automatic Difficulty Estimation of Guitar Tablatures

Feb. to May 2024

Music Technology Group, Barcelona

Collaborators: P. Ramoneda and V. Eremenko

Preparation of a new dataset and development of a new model for tablature difficulty estimation through feature engineering. Publication in preparation.

Master's Thesis - ATIAM

Automatic effect recognition and configuration for timbre reproduction

Mars-Août 2022

Sony CSL, Paris

Supervisor: Dr. Gaëtan Hadjeres

Automatic recognition and configuration of electric guitar effects from audio signals, using deep learning techniques.

Generative Audio Synthesis using a Parametric Model

Summer 2019

Digital Audio Processing Lab, Electrical Engineering Department, IIT Bombay, India

Supervisor: Pr. Preeti Rao

Using a parametric representation of musical sound signals for improved control over sound generation using a neural network (VAE).

Publications

Conditional Generation of Bass Guitar Tablature for Guitar Accompaniment [...]

2025

AI Music Creativity (AIMC)

Olivier Anoufa, [Alexandre D'Hooge](#), Ken Déguernel

[Access Link](#)

— CRediT: Conceptualization, Formal analysis, Methodology, Software, Supervision, Validation, Visualization, Writing.

What song now? Personalized Rhythm Guitar Learning in Western Popular Music

2025

International Society for Music Information Retrieval Conference (ISMIR)

Zakaria Hassein-Bey, Yohann Abbou, [Alexandre D'Hooge](#), Mathieu Giraud, Gilles Guillemain, Aurélien Jeanneau

[Access Link](#)

— CRediT: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Writing.

Vers une taxonomie et une analyse des gestes guitaristiques dans le Brutal Death Metal (FR)

2025

Journées d'Informatique Musicale (JIM)

Quentin Guilluy, Anis Farji, Joao Fernandes, Mathieu Giraud, [Alexandre D'Hooge](#), Emmanuel Leguy

[Access Link](#)

— CRediT: Methodology, Supervision, Validation, Writing (Review and Editing).

Towards Explainable and Interpretable Music Difficulty Estimation: [...]

2024

International Society for Music Information Retrieval Conference (ISMIR)

Pedro Ramoneda, Vsevolod Eremenko, [Alexandre D'Hooge](#), Emilia Parada-Cabaleiro, Xavier Serra

[Access Link](#)

— CRediT: Formal analysis, Software, Validation, Visualization, Writing.

Suggestions Pédagogiques Personnalisées pour la Guitare (in French)

2024

Journées d'Informatique Musicale (JIM)

[Alexandre D'Hooge](#), Mathieu Giraud, Yohann Abbou and Gilles Guillemain

[Access Link](#)

— CRediT: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing.

Guitar Chord Diagram Suggestion for Western Popular Music

2024

Sound and Music Computing Conference (SMC)

[Alexandre D'Hooge](#), Louis Bigo, Ken Déguernel and Nicolas Martin

[Access Link](#)

— CRediT: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing.

From MIDI to Rich Tablatures: [...]

2024

Sound and Music Computing Conference (SMC)

Pierluigi Bontempi, Daniele Manerba, [Alexandre D'Hooge](#), Sergio Canazza

[Access Link](#)

— CRediT: Data curation, Formal analysis, Software, Visualization, Writing.

Modeling Bends in Popular Music Guitar Tablatures

2023

International Society for Music Information Retrieval Conference (ISMIR)

[Alexandre D'Hooge](#), Louis Bigo and Ken Déguernel

[Access Link](#)

— CRediT: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing.

Vapar Synth: A Variational Parametric Model for Audio Synthesis

2020

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)

Krishna Subramani, Preeti Rao and [Alexandre D'Hooge](#)

[Access Link](#)

— CRediT: Software, Visualization.

Generative audio synthesis with a parametric model

2019

Late-Breaking/Demo ISMIR 2019

Krishna Subramani, [Alexandre D'Hooge](#) and Preeti Rao

[Access Link](#)

— CRediT: Conceptualization, Formal Analysis, Investigation, Software, Validation, Visualization, Writing.

Seminar and Poster Presentations

July 2025: Presentation at the IASPM 23rd International Conference on *Augmented Guitar Tablatures as pseudo-recordings*.

March 2025: Presentation for the one-day event on Music Research at Université de Lille: *L'Apprentissage Automatique appliqué aux Tablatures de Guitare en Musiques Actuelles Occidentales*.

February 2025: Presentation on *Machine Learning applied to Guitar Tablatures* at *Journées Partition et Numérique* of PEPR ICCARE.

October 2024: Poster Presentation of my work for the HCERES evaluation of the CRISAL lab: *Models for Assisted Guitar Tablature Composition*.

December 2023: Poster Presentation on Rhythm Guitar Continuation at DMRN+18: *Rhythm Guitar Tablature Continuation from Chord Progression and Tablature Prompt*.

September 2023: Talk for the MaDICS' Musiscale group: *Modéliser les tablatures pour guitare*.

Scientific Service

Since April 2024: Member of the ISMIR Ethics Working Group on Environmental Impact.

Paper Reviews

Since 2024, I have been a reviewer for the ISMIR conference, the ICASSP conference, and the LAMIR and HCMIR Workshops.

Languages

French: Native speaker

English: Proficient user

C2 Level assessed by the Cambridge Advanced Examination

Spanish: Good Command

B2 level

Computer skills

Deep Learning Programming: Pytorch and Tensorflow

Object Oriented Programming: Java, C++, Python

Other Programming languages: C, Matlab, PHP, R

Web programming: HTML, CSS, network configuration, LAMP server

Embedded electronics: Eagle, Arduino, VHDL

System administration: Bash, general Linux knowledge

Miscellaneous: $\LaTeX$, Blender, Gimp, Ardour, PureData, Faust

Personal Interests

Instrument Practice: Electric Guitar and Sousaphone.

Music Composition: French Rock and Electronic Music.

TableTop RPGs: Both as a Game-Master and a Player (Pathfinder, DnD5, Cthulhu).

Sports: Biking, Hiking, Running, Boulder-Climbing.

Handcraft: Knitting and crochet.